ARAZVERA TECHNOLOGIES / ENGINEERING PORTFOLIO / 10 NOV 2026

LEVEL 4

Enterprise Resilience & Security Engineering

Acceptance-driven system reliability, secure backend recovery and safe change engineering

Specialist Capability Catalog — verified synthetic engineering laboratory

The ARAZVERA Level 4 Control Plane is a working, disposable, **single-host laboratory**, not a customer production deployment, security certification or guarantee of uptime. All evidence below comes from bounded, isolated tests with separately documented exclusions.

Why Level 4 engineering matters

Complex backends often fail at the boundaries between APIs, identity systems, transactions, independent workers, monitoring and release control. This offering focuses on reproducible failure cases, measured recovery, explicit authorization, conservative release gates, and customer-specific acceptance criteria.

Six specialist service families

Specialist service	Representative authorized engineering work
Architecture & Reliability Audit	Dependencies, operational risk mapping, incident pathways, SLO gaps and remediation acceptance plans
Multi-Tenant Security Hardening	Restricted PostgreSQL roles, FORCE RLS, RS256/JWKS, tenant-bound scopes, invalid-token and replay testing
Transaction & Recovery Engineering	Transactional outbox, idempotent keys, lease fencing, retry/dead-letter strategies, worker recovery, logical restore
Release & Rollback Engineering	Measured synthetic Canary routes, failing-candidate HOLD, approved staging rollback and explicit evidence
Incident & SLO Readiness	Measured service probes, durable incident history, queue/latency metrics and controlled operational evidence
Backend Reliability Remediation	Python, FastAPI, PostgreSQL and Docker integration fixes with real regression tests and documented handover

Verified technical evidence

Every number below reflects an **executed** local test or GitHub Actions job. Groups overlap; never add these counts as if they were unique test cases.

Verified activity	Actual result	Boundary
Private GitHub Actions	**4 / 4 jobs PASSED**	Commit **789a044**, run **42766121925**; Python 3.11/3.12, PostgreSQL and disposable multi-service Docker
Full local Docker Pytest	**95 passed, 0 failed**	Two non-blocking Pytest cache warnings; signed identity, RLS, Worker, recovery and Canary
Local Python unit tests	**73 passed**	Security, APIs, fail-closed policies, environment and fault preflight
Selenium / Chrome UI	**13 / 13 passed**	Real dashboard probes, release HOLD, healthy/faulty Canary and rollback
Real synthetic Canary	**3 / 3 passed**	200 measured candidate HTTP requests; failing candidate produces HOLD and Stable routing
Scoped API + PostgreSQL outages	**17 / 17 passed**	Actual stop/start of labelled lab-only containers; no false healthy status
Logical PostgreSQL restore	**8 / 8 passed**	Second disposable PostgreSQL 16 container; checksum, outbox, release history and RLS
Worker stop/restart	**7 / 7 passed**	Durable pending event delivered and incident resolved after recovery

Technical architecture

The isolated Level 4 Docker project contains a FastAPI Control Plane and authenticated English dashboard, PostgreSQL 16, independent event Worker, periodic health observer, disposable downstream service, ephemeral RS256 identity provider and separate Stable/Candidate version containers. The operator API is exposed only on the Windows loopback interface. Synthetic backend and identity services have no published host ports.

Safe Canary: A local operator-approved test sends 200 actual HTTP requests to the internal candidate, measures status and latency and writes a durable decision record. A failing or unsafe candidate produces HOLD and automatically restores the **synthetic Stable route**. Manual lab-only rollback is independently auditable.

Recovery: API and dedicated PostgreSQL stop/start experiments verified unavailable/timeout outcomes, preservation of existing records and bounded recovery. A transport timeout is recorded as unknown/unhealthy, not misrepresented as a successful HTTP response.

Engagement model and safety limits

Acceptance-based delivery

Stage	Technical deliverable
1. Assess	Document scope, customer-authorized infrastructure, reproduced failures, risk and acceptance boundaries
2. Reproduce	Isolated baseline, negative paths, observable telemetry and controlled fault experiment
3. Engineer	Reviewable changes to backend, data, identity, Worker, monitoring or staging release design
4. Verify	Integration and security tests, recovery/canary evidence and explicit recorded limitations
5. Handover	Engineering report, English runbook, change/rollback conditions and acceptance register

Projects can be scoped as written milestones or time-and-materials. Third-party infrastructure, credentials and external APIs are client-owned and separately authorized. No price, completion date, existing Enterprise client history or commercial outcome is guaranteed.

Exclusions that remain explicit

- No proven deployment or availability guarantee in a customer's Enterprise production environment.
- The synthetic RS256 provider and scoped tenant roles do not establish integration with a customer's IAM/SSO, PKI or production RBAC.
- Single-host logical PostgreSQL restoration does **not** prove multi-host disaster recovery, offsite backups or measured production RPO/RTO.
- Canary/rollback affects only lab-only **HTTP staging routing**, not a real customer image deployment or production release.
- SHA-256 manifests are content checksums, **not** an operator-authenticated signature or independently issued certificate.
- A passing local/hosted test suite and basic secret pattern scan do not replace independent penetration tests, complete SBOM review or signed software provenance.
- No confidential source, keys, customer information, third-party endpoints, financial services or unrelated computer resources are included in this publication.

Contact and public resources

ARAZVERA TECHNOLOGIES

Company GitHub

Engineering Portfolio

Level 4 Public Summary

Request a scoped technical assessment with sanitized architecture, one reproducible failure scenario and proposed acceptance criteria.

This document is a verified laboratory capability statement, not a customer production certification or a representation of previously completed Enterprise contracts.